



Spectral model and experimental validation of hysteretic and aerodynamic damping in dynamic analysis of overhead transmission conductor



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ABSTRACT

The paper treats a new approach of dynamic analysis of a conductor cable of an overhead transmission line under the theoretical background of spectral element method (SEM). The methodology relies on the analytical solution of the displacement wave equation in the frequency domain; moreover, it both enhances the accuracy of model predictions and reduces the computational efforts when compared to a finite element (FE) model. Two numerical models based on SEM are built for transmission lines taking into account hysteretic and aerodynamics damping and whose analyses consider dispersion diagrams and frequency response functions (FRFs). As SEM leads to a transcendental eigenvalue problem, to obtain the natural frequencies of the conductor, it is used the Wittrick–Williams algorithm. The results presented in the paper show the sensitivity of the response conductor changes according to the tensile load and damping parameters. Finally, the numerical models have compared with the analytical solution of the cable and experimental tests performed at CEPEL's (Electric Power Research Center) laboratory with the overhead conductor Grosbeak cable. The results show outstanding accuracy in both experimental measurements and analytical analysis aspects.

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1. Introduction

Overhead transmission lines are subjected continuously to variable wind loads, which may gradually lead to the impairment of their durability, resulting in shortened service life. In this way, the design and construction of the overhead transmission lines considering a wide range of load cases acting on these slender structures are needed. Wind forces are the main cause of the conductor vibration [1], what eventually may lead to galloping, wake, or aeolian vibrations. Generally, the galloping occurs in a frequency from 0.1 to 1 Hz with amplitudes from 0.1 to 1 of conductor sag. Wake induced vibrations results in a frequency from 0.15 to 10 Hz and amplitudes from 0.5 to 80 times the conductor diameter [2,3]. As for the aeolian vibrations, they are caused by lift forces arising from the periodic shedding of vortices in the wake of the conductor and happened in a frequency from 3 to 150 Hz with amplitudes lower than the conductor diameter.ultima versão Because of the

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wind flow and on the mechanical damping of the conductor, the dynamic stresses and strains induced in the constituent wires of the conductors may increase in critical points, e.g. at the suspension clamps and the attachment points of Stockbridge dampers. High amplitudes are observed in the vertical plane, whereas the frequencies depend on the type of conductor and vibrations [4]. Summing up, it can cause fatigue damages on the wires and may lead to catastrophic consequences such as the complete rupture of the conductor and interruption on the supply of electric energy [5–7].

Several researchers have been analysing the vibration and the vibration control in overhead transmission conductors. The vibration in cables due to free and forced response using substructural interactions was performed in [8], Barbieri et al. [9,10] presented theoretical and experimental studies in the dynamical behaviour of electric cables of transmission lines. McClure and Lapointe [11] summarised a macroscopic modelling approach to line dynamic analysis and the cable dynamics model using the commercial software ADINA. A review of mechanical models of cables was published by Cardou [12] and Spak et al. [13–15].

Mathematical models of the conductor that are found in the literature encompass a wide domain from the simplest ones based on taut strings without bending stiffness to more complex where a homogeneous elastic beam with constant bending stiffness, and subjected to a constant axial load are considered [5,16–19]. By considering the cable sag analysed by different numerical methods as finite difference method [20], finite element method (FEM) [21], boundary element method [22], FEM with a central difference method (CDM) [23], and the dynamic stiffness method [24]. Although the computational models found in the literature are efficient for some dynamic analysis, they may be computational time cost intensive or infeasible for specific applications such as structural health monitoring and reliability analysis. Therefore, such as wave finite element (WFE) [25,26] (applied in cable), and the spectral element method [27,28] are attractive alternative computational models.

The spectral element method has the dynamic system governing equations written in the frequency domain. It is a meshing method similar to FEM, where the approximated element shape functions are substituted by dynamic shape functions obtained from the exact solution of governing differential equations. Therefore, a single element is sufficient to model any continuous and uniform part of the structure. This feature reduces significantly the number of elements required in the structure model and improves the accuracy of the dynamic system solution. Moreover, a linear problem solution is solved by using global system matrix equation related to global spectral nodal degrees-of-freedom (dofs), in time-domain solution only is needed the inverse discrete Fourier transform (DFT). The advantages of the SEM are low computational costs, effectiveness in dealing with frequency-domain problems and with the non-reflecting boundary conditions of the infinite or semi-infinite-domain problems [28]. The SEM [29,27,28] has been applied in many applications, as the study of the rotor dynamics, composite laminated, and periodic lattice performed by Lee [28], the wave behaviour in composites and inhomogeneous media with applications to structural health monitoring and active vibration control [30,31], and related to structural damage detection including stochastic and wave propagation [32–35]. The spectral analysis of an undamaged beam considering axial tensile load were approached by Doyle [27] and Gopalakrishnan [36]. The spectral element for the Timoshenko beam subjected to a moving axial tension is modelled was presented by Lee et al. [37]. Choi and Inman [38] modelled a cable-harnessed structure employing the SEM, and Zang et al. [39] proposed an undamped Timoshenko beam with axial tension combined with FEM and a wave propagation analysis in the cable structure. Dutkiewicz and Machado [40,50] studied the vibrations of overhead transmission lines and the theoretical background for an undamped system, and in [41] the authors treated the dynamic response of overhead transmission line under the effect of turbulent wind flow, both using the numerical model based on the SEM.

The main goal of this paper is to develop two numerical models to formulate the overhead transmission conductor by considering hysteretic and aerodynamic dampings using the spectral element method. It is well known that the damping is essential information in a dynamic system analysis [42]; the numerical models with damping using the SEM have not been found in the literature. Thus, objectives of the present study are to address three major issues: (i) considering the SEM with hysteretic damping into the model formulation; (ii) integrating the aerodynamic and friction damping into the dynamic model and develop a spectral element; and (iii) demonstrating the efficiency of the two numerical model and validate then by comparing with experimental data, and with the analytical solution. The natural frequencies of the spectral conductor model are obtained by using the Wittrick-Williams algorithm to compute all required natural frequencies efficiently.

2. Spectral element method for a conductor cable

The governing equation of the undamped vibration of cables is equivalent to the one of the Euler-Bernoulli beams subjected to a tensile load [43,44],

$$EI \frac{\partial^4 v(x,t)}{\partial x^4} - T \frac{\partial^2 v(x,t)}{\partial x^2} = \rho A \frac{\partial^2 v(x,t)}{\partial t^2} \quad (1)$$

where ρA is mass per unit length, EI the uniform bending rigidity, L cable length, T is tension force, and $v(x,t)$ is the cable displacement as a function of the position x and time t . As for the natural frequencies, they can be obtained analytically as follows

$$\omega_n = \frac{\pi^2}{L^2} \sqrt{\frac{EI}{\rho A} \left(n^2 + \frac{n^2 T L^2}{\pi^2 EI} \right)}^{1/2} \quad n = 1, 2, \dots \quad (2)$$

The Euler–Bernoulli beam equation of motion subjected to a tensile load and under bending vibration is also described based in Eq. (1). As for the structural internal damping, it is introduced into the beam formulation by considering that the relation between stress and strain in the frequency domain is given by a complex modulus $E = E_0(1 + \eta i)$ where E_0 is the nominal Young's modulus value, and η is the hysteretic structural loss factor. Fig. 1 shows an elastic two-node beam element with a uniform rectangular cross-section subjected to an tensile load including the nodal forces and displacement.

2.1. Spectral analysis

The analyses on wave propagation in waveguides is to understand its physical phenomenon which is described by the wave parameters, such as the wavenumber, speed wave and other features like the existence of cutoff frequencies, etc [36]. In order to obtain the wave parameters (wave number and group velocity), it is necessary to perform a spectral analysis on the governing equations of motion. The spectral form of the displacement field variables can be obtained by using the DFT for the displacement field $v(x, t)$ as follows

$$\mathbf{v}(x, t) = \sum_{n=1}^N \hat{\mathbf{v}}(x, \omega_n) e^{i\omega_n t} = \sum_{n=1}^N \left(\sum_{j=1}^4 \hat{\mathbf{v}}_j e^{-ik_j x} \right) e^{i\omega_n t}, \quad (3)$$

where $i = \sqrt{-1}$, $\omega_n = n \times \omega_0$ corresponds to a circular frequency in radians per second at n th sampling point, N is the Nyquist point in DFT, $\mathbf{v}(x, t)$ is the transverse displacement, and $\hat{\mathbf{v}}(x, \omega_n)$ is the Fourier coefficient limited from $n = 0$ to N . Assuming the solution for the displacements as $\hat{\mathbf{v}}_j$, it is suitable to calculate the wavenumbers k_j associated with the j th wave mode. By substituting Eq. (3) into Eq. (1), considering free vibration, hysteretic damping, and transforming the partial equation into an ordinary differential equation, we arrive at

$$EI \frac{d^4 \hat{v}}{dx^4} - T \frac{d^2 \hat{v}}{dx^2} - \omega^2 \rho A \hat{v} = 0 \quad (4)$$

By using the spectral form of solution as $\hat{v}_j = \hat{v}_j e^{-i(kx - \omega t)}$ in Eq. (4), we get the characteristic equation for determination of wavenumbers, which is a fourth-order characteristic polynomial equation in k_j , given by [27,36]

$$EI k^4 + T k^2 - \omega^2 \rho A = 0, \quad (5)$$

The characteristic equation is quadratic in k^2 and hence can be easily solved. There are four roots, representing two sets of wave mode pairs, being pure real roots and two pure imaginary roots in the form

$$k_1 = \pm \sqrt{-\frac{T}{2EI} + \sqrt{\left(\frac{T}{2EI}\right)^2 + \frac{\rho A \omega^2}{EI}}}, \quad k_2 = \pm \sqrt{-\frac{T}{2EI} - \sqrt{\left(\frac{T}{2EI}\right)^2 + \frac{\rho A \omega^2}{EI}}}, \quad (6)$$

The four roots allow us the complete solution. To generalize the notation, for the Euler–Bernoulli beam spectral element subjected to axial load of length L , we assumed that

$$\pm k_1 = k \quad \text{and} \quad \pm k_2 = ik$$

Thus, the spectral solution considering the travelling waves can be expressed in the form

$$\hat{v}(x, \omega) = a_1 e^{-ikx} + a_2 e^{-kx} + a_3 e^{-ik(L-x)} + a_4 e^{-k(L-x)} = \mathbf{s}(x, \omega) \mathbf{a} \quad (7)$$

Section 2.2 presents an spectral beam element formulation subjected to a tensile loads and that takes into account hysteretic damping, which is considered by including the structural loss factor into the Young's modulus. Section 2.3 presents an spectral element formulation for beam under tensile loads that takes into account both hysteretic and viscous damping.

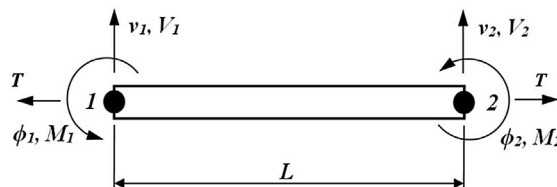


Fig. 1. Two-node beam spectral element.

2.2. Beam spectral element subjected to a tensile load with hysteretic damping

For the spectral beam element subjected to a tensile load, the general solution of Eq. (4) can be obtained as expressed in Eq. (7). Where

$$\mathbf{s}(\mathbf{x}, \omega) = \{e^{-ikx}, e^{-kx}, e^{-ik(L-x)}, e^{-k(L-x)}\},$$

$$\mathbf{a}(\mathbf{x}, \omega) = \{a_1, a_2, a_3, a_4\}^T$$

The spectral nodal displacements and slopes of the beam element are related to the displacement field at node 1 ($x = 0$) and node 2 ($x = L$), illustrated in Fig. 1, as follows

$$\mathbf{d} = \begin{bmatrix} v_1 \\ \phi_1 \\ v_2 \\ \phi_2 \end{bmatrix} = \begin{bmatrix} v(0) \\ v'(0) \\ v(L) \\ v'(L) \end{bmatrix} \quad (8)$$

By substituting Eq. (7) into the right-hand side of Eq. (8) allows one to rewrite it in matrix form as follows

$$\mathbf{d} = \begin{bmatrix} \mathbf{s}(0, \omega) \\ \mathbf{s}'(0, \omega) \\ \mathbf{s}(L, \omega) \\ \mathbf{s}'(L, \omega) \end{bmatrix} \mathbf{a} = \mathbf{G}_B(\omega) \mathbf{a} \quad (9)$$

where

$$\mathbf{G}_B(\omega) = \begin{bmatrix} 1 & 1 & e^{-ikL} & e^{-kL} \\ -ik & -k & ie^{-ikL} & e^{-kL} \\ e^{-ikL} & e^{-kL} & 1 & 1 \\ -ie^{-ikL} & -e^{-kL} & ik & k \end{bmatrix} \quad (10)$$

The frequency-dependent displacement within an element is interpolated from the nodal displacement vector $\mathbf{d}$ by eliminating the constant vector $\mathbf{a}$ from Eq.(8)using Eq.(9) it is expressed as

$$\hat{\mathbf{v}}(\mathbf{x}, \omega) = \mathbf{g}(\mathbf{x}, \omega) \mathbf{d} \quad (11)$$

where the shape function is

$$\mathbf{g}(\mathbf{x}, \omega) = \mathbf{s}(\mathbf{x}, \omega) \mathbf{G}_B^{-1}(\omega) = \mathbf{s}(\mathbf{x}, \omega) \Gamma(\omega) = \begin{Bmatrix} g_1(\mathbf{x}, \omega) \\ g_2(\mathbf{x}, \omega) \\ g_3(\mathbf{x}, \omega) \\ g_4(\mathbf{x}, \omega) \end{Bmatrix} \quad (12)$$

$$= \begin{Bmatrix} \frac{-2 \cos(kx) - 2 \cosh(kx) + (1-i)(\cos(k((1+i)L-x)) + i \cos(k((1+i)L-ix)) + \cosh(k((1+i)L-x)) + i \cosh(k((1+i)L-ix)))}{4 \cos(kL) \cosh(kL)} \\ \frac{-2 \sin(kx) + 2 \sinh(kx) + (1+i)(\sin(k((1+i)L-x)) - \sin(k((1+i)L-ix)) + \sinh(k((1+i)L-x)) - \sinh(k((1+i)L-ix)))}{4k(\cos(kL) \cosh(kL) - 1)} \\ \frac{\cos(k(L-x)) - \cos(kx) \cosh(kL) + \cosh(k(L-x)) - \cos(kL) \cosh(kx) + \sin(kx) \sinh(kL) - \sin(kL) \sinh(kx)}{2 - 2 \cos(kL) \cosh(kL)} \\ \frac{\sin(k(L-x)) - \cos(kx) \sinh(kL) + \cosh(kx)(\sinh(kL) - \sin(kL)) + \cosh(kL)(\sin(kx) - \sinh(kx)) + \cos(kL) \sinh(kx)}{2k(\cos(kL) \cosh(kL) - 1)} \end{Bmatrix}$$

In the case of the Euler-Bernoulli beam, a generalized transverse displacement at an arbitrary point x can be expressed as,

$$\hat{v}(x) = g_1(x) \hat{v}_1 + g_2(x) \hat{\phi}_1 + g_3(x) \hat{v}_2 + g_4(x) \hat{\phi}_2 \quad (13)$$

One of the advantages of using SEM is that only one element is required for a homogeneous structural member. The global dynamic spectral matrix for a system can be described as

$$\mathbf{S}(\omega) = \mathbf{K}(\omega) - \omega^2 \mathbf{M}(\omega) \quad (14)$$

The dynamic stiffness matrix $\mathbf{K}(\omega)$ for the two-node spectral beam element subject to tensile load T expressed in Eq.(14), has the mass and stiffness matrices determined in a weak form by:

$$\mathbf{K}(\omega) = \int_0^L \left(E I \mathbf{g}''(x)^T \mathbf{g}''(x) + T \mathbf{g}'(x)^T \mathbf{g}'(x) \right) dx \quad (15)$$

and

$$\mathbf{M}(\omega) = \rho A \int_0^L \mathbf{g}(x)^\top \mathbf{g}(x) dx \quad (16)$$

where $(\bullet)'$ express the spatial derivative. As far as the structure beam is uniform without any sources of discontinuity, it can be represented by a single spectral element with very accurate solutions[27]. However, in case some source of discontinuity is present the beam should be discretized into spectral elements. Analogous to FEM [45], the spectral elements can be assembled to form a global structure matrix system [28].

2.3. Beam spectral element subjected to a tensile load with aerodynamic and friction damping

As presented in the last section the cable was modelled as an equivalent homogeneous Euler-Bernoulli beam with constant bending stiffness, subjected to a tensile load with hysteretic damping which was included into the Young's modulus. Some other possibility to take the system dissipation into account is to consider aerodynamic damping. In this context, Matt and Castello[19,46] presented a model that takes into account aerodynamic damping and the inner dissipation mechanisms are modelled by means of a Kelvin-Voigt constitutive equation as described by the following:

$$EI \frac{\partial^4 v}{\partial x^4} - T \frac{\partial^2 v}{\partial x^2} + \xi I \frac{\partial}{\partial t} \left(\frac{\partial^4 v}{\partial x^4} \right) + \alpha \frac{\partial v(x, t)}{\partial t} + \rho A \frac{\partial^2 v(x, t)}{\partial t^2} = q(x, t) \quad (17)$$

where the parameter α is the equivalent aerodynamic damping coefficient, E is the Young modulus, and ξI represents the energy dissipation mechanism associated with the inter-strand friction among the wires of a typical conductor, ξ being a material damping factor. One advantage when using this model is that the inclusion of aerodynamic damping may naturally use this approach are the inclusion of the aerodynamic damping may naturally be coupled with the fluid dynamics equations in order to simulate the fluid-structure interaction problem governing the aeolian vibrations[47,19]. The equilibrium Eq.(17) in the frequency domain can be written as

$$(EI + i\omega\xi I) \frac{d^4 \hat{v}}{dx^4} - T \frac{d^2 \hat{v}}{dx^2} + (i\omega\alpha - \omega^2 \rho A) \hat{v} = \hat{q} \quad (18)$$

In order to investigate the wave propagation characteristics in the cable described by Eq. (18) one firstly considers the solution $v(x, t) = \hat{v}e^{-i(kx - \omega t)}$ when $\hat{q} = 0$ and obtains the dispersion equation as follows

$$(EI + i\omega\xi I)k^4 + Tk^2 + i\omega\alpha - \omega^2 \rho A = 0, \quad (19)$$

By solving the Eq. (19) in function of the quadratic term in k^2 , we will have two distinct wave modes in the positive direction with wavenumbers expressed as

$$k_1 = \pm \sqrt{-\frac{T}{2EI + i\omega\xi I} + \sqrt{\left(\frac{T}{2EI + 2i\omega\xi I}\right)^2 - \frac{i\omega\alpha - \rho A\omega^2}{EI + i\omega\xi I}}}, \quad k_2 = \pm \sqrt{-\frac{T}{2EI + i\omega\xi I} - \sqrt{\left(\frac{T}{2EI + 2i\omega\xi I}\right)^2 - \frac{i\omega\alpha - \rho A\omega^2}{EI + i\omega\xi I}}}. \quad (20)$$

The spectral element with respective shape function in this case is formulated by applying similar procedure presented from Eqs. (7)–(12). The global damped dynamic spectral matrix system can be described as

$$\mathbf{S}(\omega) = \mathbf{K}(\omega) + i\omega\mathbf{C}(\omega) - \omega^2\mathbf{M}(\omega) \quad (21)$$

Eq. (14) expresses the damped dynamic stiffness matrix for the two-node spectral beam element, where the damping matrix $\mathbf{C}(\omega)$ is assumed as proportional to the mass and stiffness. The mass and stiffness matrices derived in a weak form are:

$$\mathbf{K}(\omega) = \int_0^L \left(EI \mathbf{g}''(x)^\top \mathbf{g}''(x) + T \mathbf{g}'(x)^\top \mathbf{g}'(x) \right) dx \quad (22)$$

and

$$\mathbf{M}(\omega) = \rho A \int_0^L \mathbf{g}(x)^\top \mathbf{g}(x) dx \quad (23)$$

The elemental damping matrix contain the internal and aerodynamic damping can be rewritten as

$$\mathbf{C}(\omega) = \int_0^L \left(\xi I \mathbf{g}''(x)^\top \mathbf{g}''(x) + \alpha \mathbf{g}(x)^\top \mathbf{g}(x) \right) dx \quad (24)$$

or as function of the elemental mass and stiffness matrices as follow:

$$\begin{aligned} \mathbf{C}(\omega) &= \frac{\alpha}{\rho A} \mathbf{M}(\omega) + \int_0^L \xi \mathbf{I} \mathbf{g}''(x)^\top \mathbf{g}'(x) dx \\ &= \frac{\alpha}{\rho A} \mathbf{M}(\omega) + \frac{\xi I}{EI} \mathbf{K}(\omega) - \frac{\xi IT}{EI} \int_0^L \mathbf{g}'(x)^\top \mathbf{g}'(x) dx \end{aligned} \quad (25)$$

Considering that all parameters and matrices of Eqs. (12) ((22)–(25)) are presented, it is easy to implement them in software similar to MATHEMATICA® to obtain the shape function, stiffness, mass and damping matrices. In case the solution of the eigenvalue problems is needed, it requires a proper method to be solved because SEM leads to a transcendental eigenvalue problem. Section 3 introduces the Wittrick-Williams algorithm, the method we used to treat the eigenvalue solution.

3. Wittrick-Williams algorithm

The SEM leads to a transcendental eigenvalue problem because the global dynamic stiffness matrix is a transcendental function of frequency ω . The numerous eigensolvers available for the linear eigenvalue problems are no longer applicable in this case and an alternative is the Wittrick-Williams algorithm [48], which is an efficient technique to avoid the possibility of missing any eigenfrequencies in the searching process. The computation of the eigenfrequencies of the dynamic stiffnesses $\mathbf{S}(\omega)$ described by Eqs. (14) and (21) requires a particular procedure. Firstly one varies ω in small steps. Secondly, one may compute the determinant $\det[\mathbf{S}(\omega)]$ at each ω aiming at identifying all frequencies $\omega = \Omega$ such that $[\mathbf{S}(\Omega)] = 0$.

The Wittrick-Williams algorithm was developed to compute the natural frequencies of an undamped linear elastic system, however, it is still valid as a guide or lightly damped systems [28]. This technique specifies the total number of natural frequencies of a specific structure chosen a frequency ω , it is given by

$$J(\omega) = J_0(\omega) + \text{sgn}[\mathbf{S}(\omega)] \quad (26)$$

where $J(\omega)$ is the total number of natural frequencies of a system which are less than a chosen frequency ω , $J_0(\omega)$ is the total number of natural frequencies which would still be exceeded by the selected frequency (ω); and $\text{sgn}[\mathbf{S}(\omega)]$ is the sign count of matrix $[\mathbf{S}]$, which is equal to the number of negative elements on the diagonal of the upper triangular form of $[\mathbf{S}]$. The upper triangular form can be reduced by using the simple Gaussian elimination procedure.

The total number of natural frequencies, $J_0(\omega)$, is defined by the boundary condition. To compare the numerical model and analytical solution presented in Eq. (2) the boundary conditions are assumed as simply supported. Thus, $J_0(\omega)$ the can be calculated as

$$J_0(\omega) = J_{ss}(\omega) - \text{sgn}[\mathbf{S}_{ss}(\omega)] \quad (27)$$

For the evaluation of $J_{ss}(\omega)$, when the condition $\det[\mathbf{S}_{ss}(\omega)] = 0$, it has

$$J_{ss}(\omega) = \sum_{j=1}^m J_{ssj}(\omega) \quad (28)$$

where $J_{ssj}(\omega)$ is the number of natural frequencies that are less than (ω) for the j th structure member with supported boundaries.

4. Numerical results

This section is devoted both to compare the numerical predictions with analytical results and to perform some local sensitivity analysis of the model. All the analyses along this section consider the ACSR Grosbeak cable simply supported at both ends and whose properties are [49]: diameter of $D = 25.15$ mm, mass per unit length of $\rho A = 1.3027$ kg/m, flexural rigidity of $EI = 484$ Nm², and the span length of $L = 51.950$ m. The analytical natural frequencies shown in Eq. (2) are compared to the ones provided by the SEM model. As the SEM demands the solution of a transcendental eigenvalue problem, the Wittrick-Williams algorithm is used to compute the eigenfrequencies of Eqs. (14) and (21).

Firstly it considered the cable with a tension $T = 1670 \times 10^4$ Kg. Table 1 presents the analytical natural frequencies. The upper graph in Fig. 2 present the FRF of the system unitary force excitation and measurement located at $x = 1.606$ m, point related to shaker-Ac4 shown in Fig. 10. The lower graph in Fig. 2 presents $J(\omega)$. The frequencies at which the function $J(\omega)$ presents discrete unitary jumps corresponds the FRF peaks.

Table 1

Analytical natural frequency calculated with Eq. (2) for the tensile load of $T = 1670 \times 10^4$ kgf.

Natural Frequency [Hz]	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	ω_8
	1.1140	2.2293	3.3469	4.4681	5.5940	6.7258	7.8648	9.0119

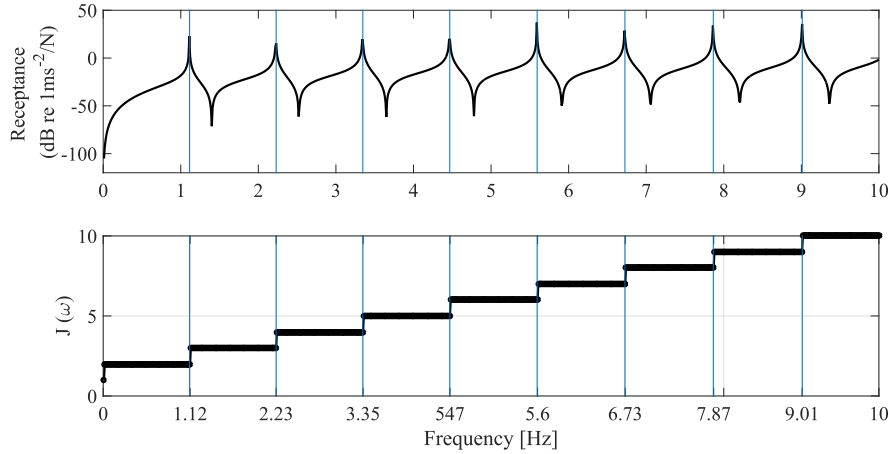


Fig. 2. Natural frequency identification using the Witrick-Williams algorithm.

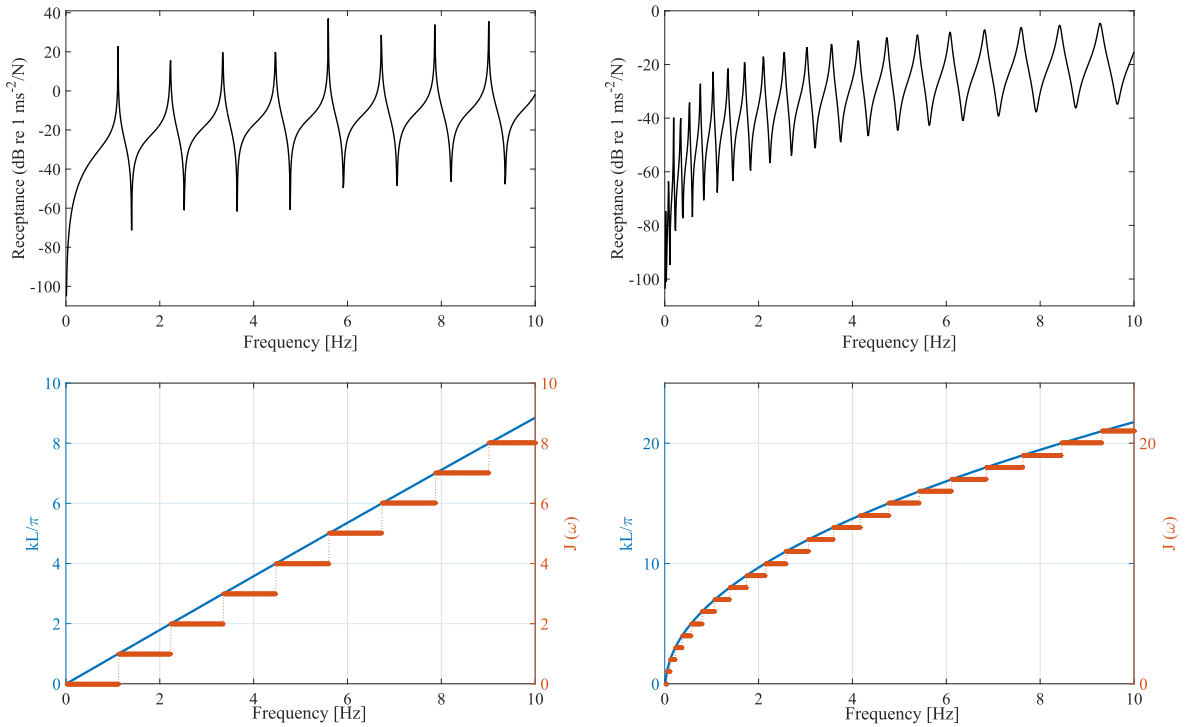


Fig. 3. FRFs, dispersion curves and the number of natural frequencies $J(\omega)$, for tensile load of $T = 1670 \times 10^4$ kgf (LHS) and $T = 1670$ kgf (RHS).

The upper graphs in Fig. 3 presents the FRFs for tensile loads $T = 1670 \times 10^4$ kgf and $T = 1670$ kgf. Further, the lower graphs in Fig. 3 present both the dispersion curve and the function $J(\omega)$. For the higher tensile load, the dispersion diagram of the conductor presents a straight line trend to an axial mode propagation generating a non-dispersive wave. The natural frequencies provided by $J(\omega)$ are in total eight, as shown in Fig. 3-(LHS). By reducing the tensile load to $T = 1670$ kgf, one identifies that the cable acts as a dispersive medium, changing to flexural wave mode propagation creating dispersive waves. In this case the number of natural frequencies increased by twenty-one, illustrated in Fig. 3-(RHS).

The dispersion diagram for both modes calculated in Eq. (6) are shown in Fig. 4. The wavenumber $+k_1$ and $+k_2$, plotted in the blue line, is the wave mode 1 and the wave solution (propagating modes); while $-k_1$ and $-k_2$, printed black dashed line, is the wave mode 2 (evanescent) predominantly spatially damped vibrations [27]. Next, it is presented some local sensitivity analysis of the FRF and the dispersion diagram concerning the aerodynamic damping coefficient α , the inner damping factor ξI , the hysteretic damping factor η and the tensile load T .

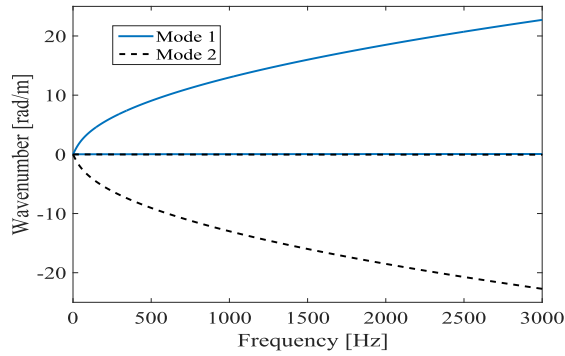


Fig. 4. Dispersion relation: Blue continuous line is real part of the wavenumber or mode 1, and black dashed line is the imaginary part or mode 2.

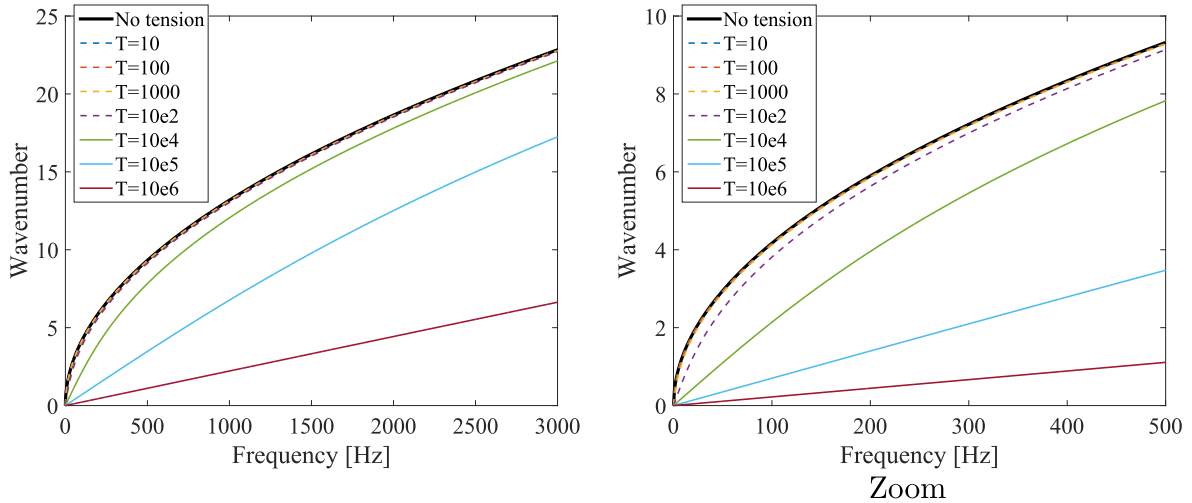


Fig. 5. Dispersion diagram obtained by the hysteretic damping model with $\xi = 0.01$ and different tension load values. Full dispersion diagram from 0 to 3000 Hz on the LHS, and on the RHS a zoom in the frequency range between 0 and 500 Hz.

Fig. 5 presents the dispersion diagram for the conductor model with hysteretic damping calculated with Eq. (6) and different values of tensile load. The wavenumber without tension is used as a reference when making comparisons considering estimates with tensile loads of $T = [10, 100, 1000, 10^3, 10^4, 10^5, 10^6]$ kgf. Likewise, **Fig. 6** shows the comparison of the dispersion diagram for the conductor model with aerodynamic damping obtained through Eq. (20) with $\alpha = 0.05$ and using the same tensile loads as hysteretic damping case. For both models, as the tension T increases, the modes move from non-dispersive ones when $T = 10^4$ to dispersive ones when up to $T = 10^3$. The predicted dispersion curves shown in **Figs. 3, 5 and 6** present two kinds of the wave propagation modes independent of the numerical model. Further, these modes are dependent on the tensile load T . Next, the influence of the damping model parameters are assessed.

Figs. 7 and 8 show the dispersion diagram of the conductor estimated with a tensile load of $T = 1670$ Kgf considering different loss factors ξ and aerodynamics damping factors α . **Fig. 7** presents the variation of the dispersion curves with respect to α when $\xi = 10^{-5}$. **Fig. 8** presents the variation of the dispersion curve with respect to ξ when $\alpha = 0.05$.

As for the variation of the FRFs with respect to damping model parameters, these analyses are shown in **Fig. 9**. **Figs. 9-a and 9-b** present the variations of the FRFs with respect to the aerodynamic damping parameter α and to the inner damping factor ξ , respectively. From these figures one may see a shift in the natural frequencies and in the peak magnitudes. By fixing the structure loss factor values of $\xi = 10^{-5}$ as for the hysteretic damping, **Fig. 9-c** presents the FRFs obtained for four values of hysteretic damping model. In this case, there is no resonance pick shift the changes occur at the amplitude of the peaks within the frequency band. For a small loss factor value $\eta = 0.001$ the resonance peaks are sharp, however with the increase of η the resonance peak tends to widen and decrease of amplitude.

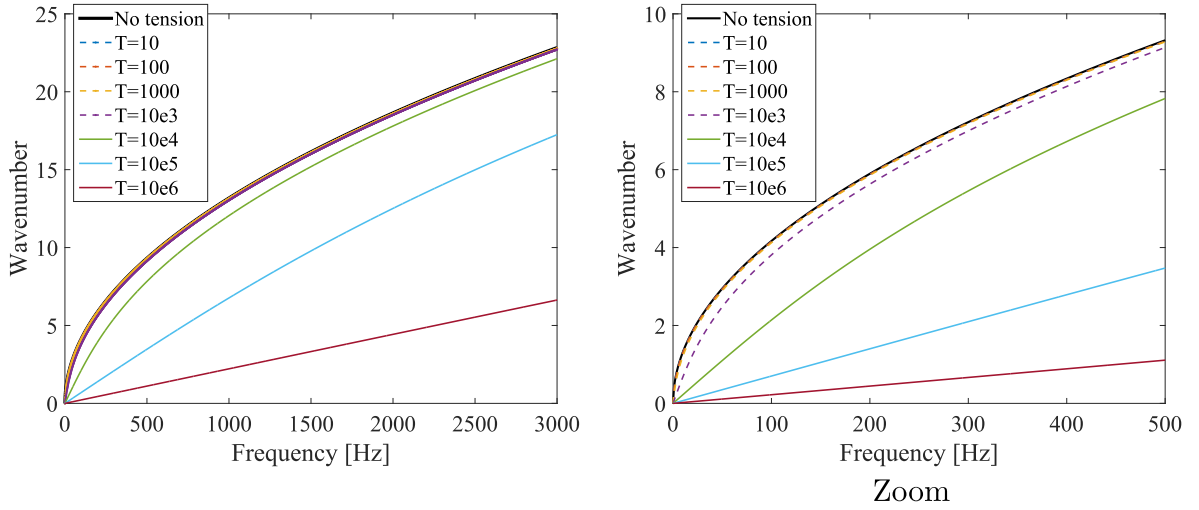


Fig. 6. Dispersion diagram obtained by the aerodynamic damping model with $\alpha = 0.05$ and different tension load values. Full dispersion diagram from 0 to 3000 Hz on the LHS, and on the RHS a zoom in the frequency range between 0 and 500 Hz.

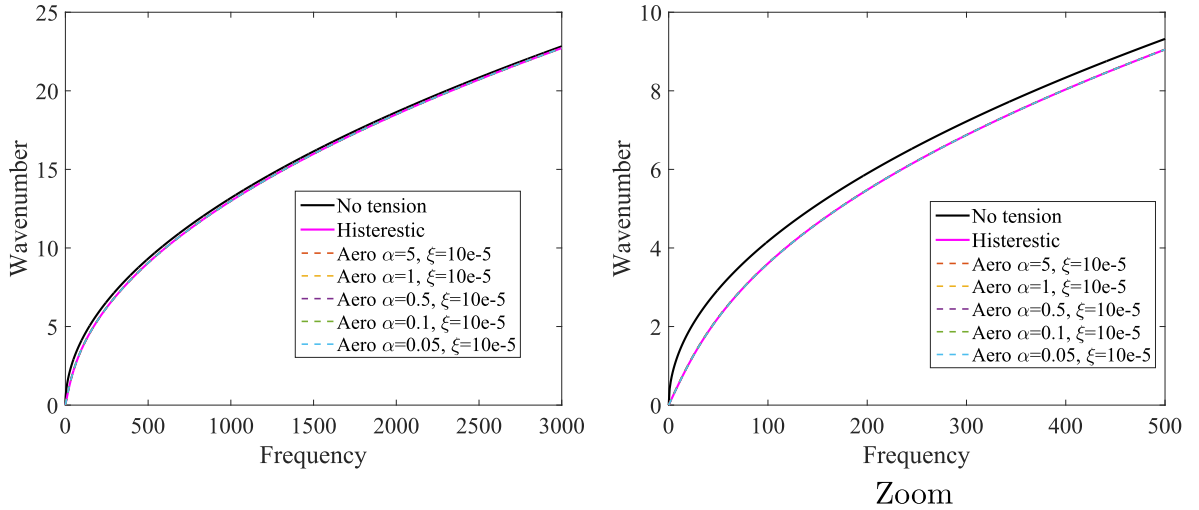


Fig. 7. Dispersion diagram obtained by the aerodynamic damping model with a fixed tensile load of $T = 1.670 \times 10^4$ kgf and different values of α . Full dispersion diagram from 0 to 3000 Hz on the LHS, and on the RHS a zoom in the frequency range between 0 and 500 Hz.

5. Experimental set-up and results

The experimental tests were performed at the laboratory span of the Electric Power Research Center (CEPEL). The transmission line conductor under analysis is the ACSR Grosbeak, that has a nominal diameter of $D = 25.15$ mm, the weight per unit length of $\rho A = 1.3027$ kg/m, flexural rigidity of $EI = 484$ Nm², and the span length of $L = 51.950$ m. The tests assumed a tensile load of 16700 N (1670 kgf) which is approximately 19% of the Grosbeak rated tensile strength (RTS), a usual percentage employed in the real system. The Grosbeak conductor is excited by an electrodynamic shaker at 1.606 m from the support and measured with four accelerometers namely Ac1, Ac2, Ac3 and Ac4, disposed at $L/2$, $L/2$ and $L/4$ m, respectively. The electrodynamic shaker was manufactured by Data Physics, model S-150 with controller DP-V150 and amplifier A-10C-05. The maximum velocity is 1.5 m/s; the maximum acceleration is 72 g; the nominal forces are 1000 N, 650 N and 1300 N respectively for sinusoidal, random and shock; and, finally, the frequency range is 2 Hz to 5 kHz. The power of the amplifier is 1250 W; the maximum shaker displacement is 25.4 mm peak-to-peak. The force transducer used is a Bruel & kjaer-model 8230-002 with nominal sensitivity of 2.41 mV/N. The accelerometers were IEPE Bruel & kjaer-model Deltatron 4519-001 with 1 gr mass.

Fig. 10 shows the experimental set up on the left and a sketch of the experiment rigs on the right. It is worthwhile to mention that the electrodynamic shaker is located at the same position as the accelerometer Ac4, which is the driving point

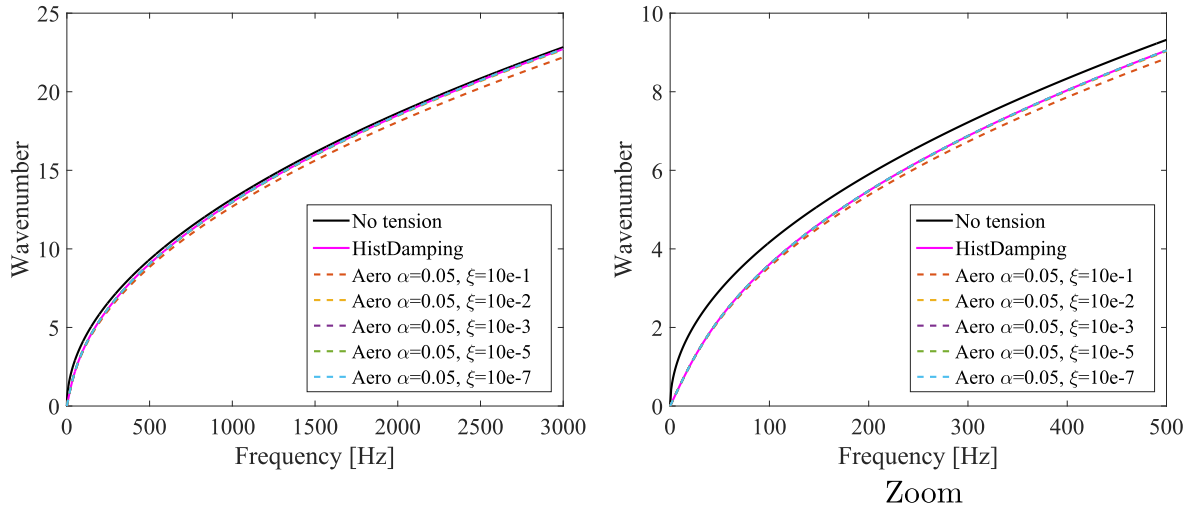


Fig. 8. Dispersion diagram obtained by the aerodynamic damping model with a fixed tensile load of $T = 1.670 \times 10^4$ kgf and different values of ξ . Full dispersion diagram from 0 to 3000 Hz on the LHS, and on the RHS a zoom in the frequency range between 0 and 500 Hz.

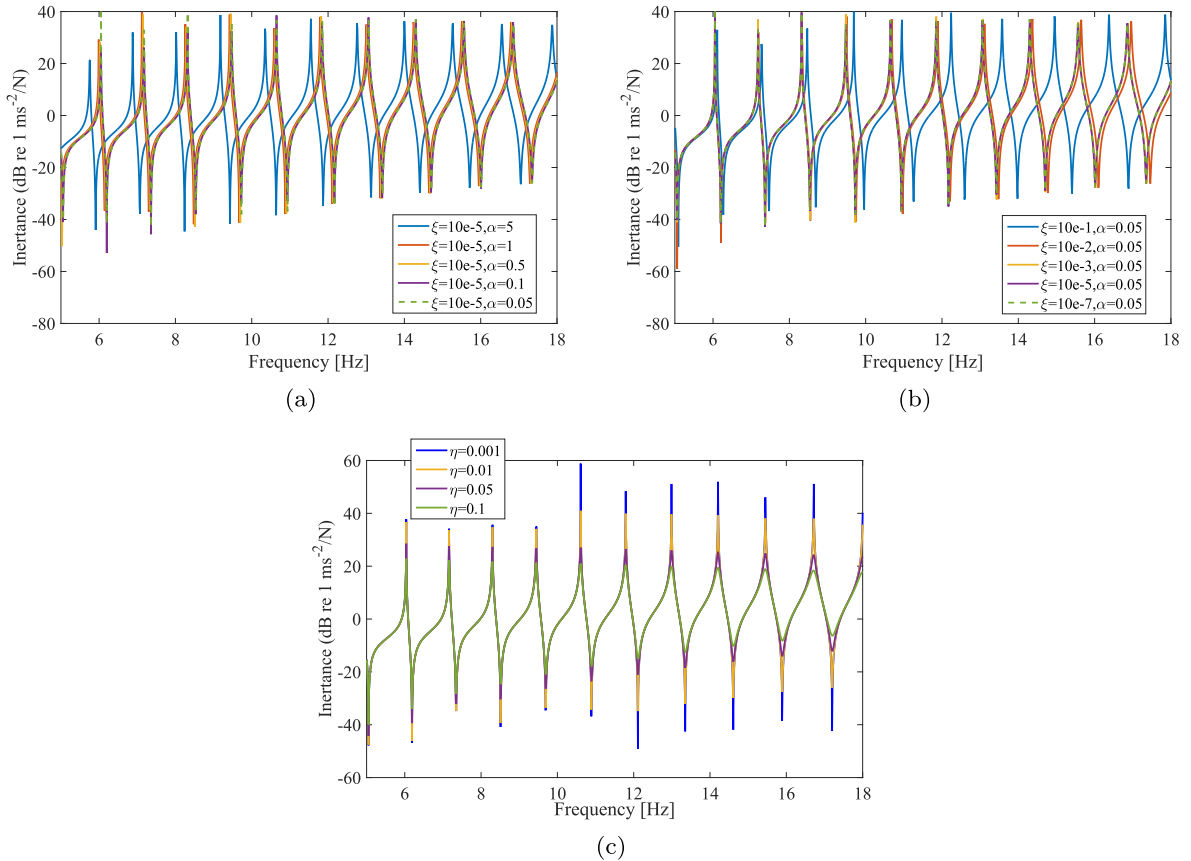


Fig. 9. Effect of damping model in FRFs: a) For the structure loss factor $\xi = 10^{-5}$ and varying the aerodynamic damping as $\alpha = [5, 1, 0.5, 0.1, 0.05]$; b) for a fixed $\alpha = 0.05$ and different values of $\xi = [10^{-1}, 10^{-2}, 10^{-3}, 10^{-5}, 10^{-7}]$; c) structural damping with loss factors of $\eta = [0.001, 0.01, 0.05, 0.1]$.

of the conductor. The signals from the electrodynamic shaker and the four accelerometers are read and recorded by PULSE acquisition data system from Bruel & kjaer, which also computes the frequency response functions directly.

Numerical results obtained with the model considering only hysteretic damping theory are compared with experimental data in Fig. 11. Fig. 12 shows the FRFs calculated with model whose dissipation is described by the viscous damping

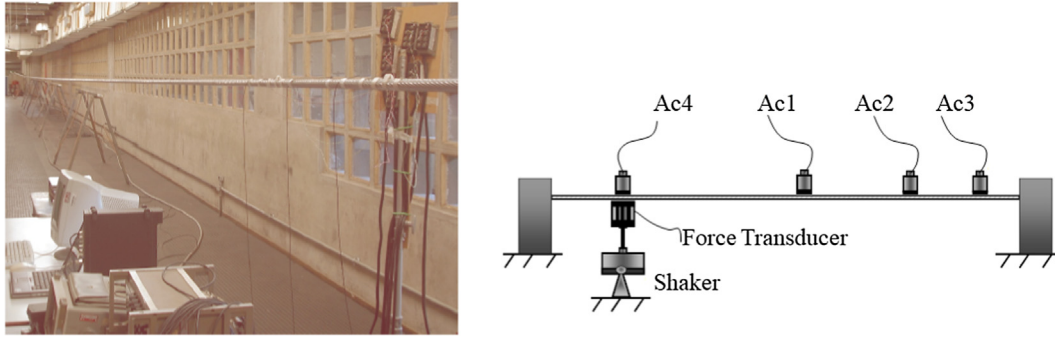


Fig. 10. Experimental set-up: general overview of the CEPEL's laboratory span (left) and schematic experimental set-up(right).

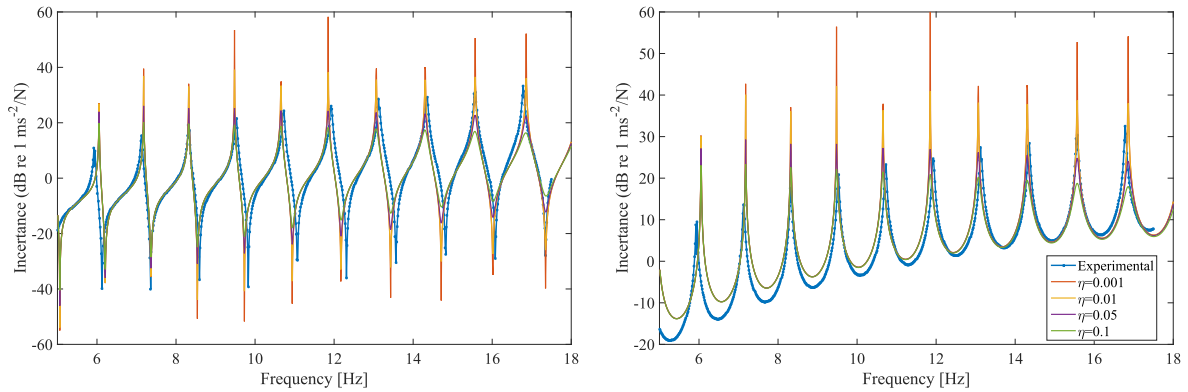


Fig. 11. Numerical FRFs calculated by the hysteretic damping model with different damping factor compared with experimental FRFs. FRFs obtained at the excitation point with accelerometer Ac4 (LHS) and at the point of the accelerometer Ac3. The tensile load of $T = 1.670 \times 10^4$ Kg.

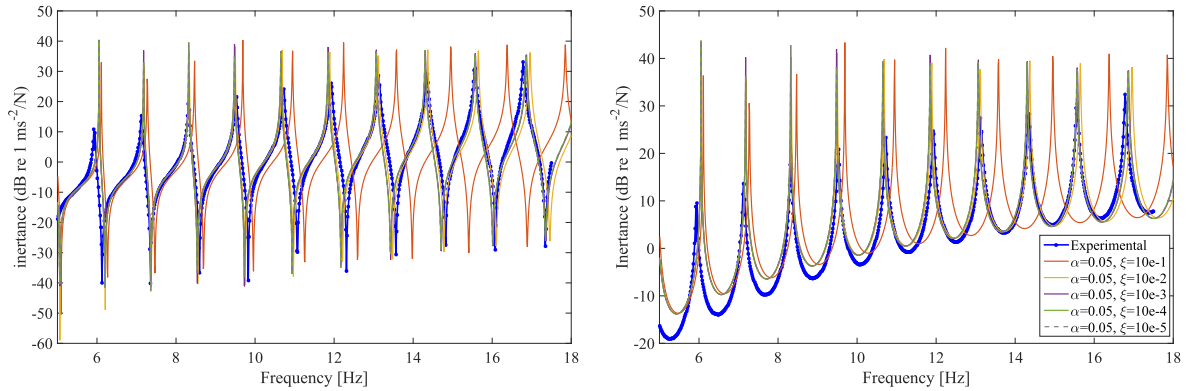


Fig. 12. Numerical FRFs calculated by the aerodynamic damping model with different damping factor compared with experimental FRFs. FRFs obtained at the excitation point with accelerometer Ac4 (LHS) and at the point of the accelerometer Ac3. The tensile load of $T = 1.670 \times 10^4$ Kg.

parameter α and the inner parameter ξ . In both cases, the measured responses are from the accelerometer Ac4 and Ac3, which correspond to the excitation point and to the accelerometer closer to the support as shown in Fig. 10.

The experimental measured FRFs are compared with the numerical model considering the hysteretic damping calculated with four values of loss factor, $\eta = [0.1, 0.05, 0.01, 0.001]$, as shown in Fig. 11. The typical values of structural loss factor found in literature ranges between 0.01 and 0.05. For values lower than 0.01 one could barely observe differences in resonance peaks. For the smallest loss factor the system does not showed big difference in the resonance peaks, as demonstrated previously. However, as the loss factor increases the changes is visible. Following the literature, the loss factors that had best approximation with the experimental FRFs were 0.01 in whole frequency range, and 0.05 from 7 to 14 Hz.

In summary, as shown above, the damping models play a significant effect on the numerical model, though both models can be used to model the conductor. The two numerical models presented good accuracy in the range of both experimental measurements, however, the aerodamping showed to be a complete model once it includes both types of damping presented in the system.

6. Conclusion

In this paper, a significant issue of vibration and spectral analysis in a single overhead transmission conductor considering two cases of damping into the numerical models were performed. The computational modes for the conductor were built based on the spectral element method. In the literature, a limited number of papers treat the problem of vibration in cables using the SEM, and the ones that approaches the problem have not considered the damping, which is important information in the dynamic of systems behaviour. Thus, it was proposed a beam spectral element under a tensile load including the structural damping and another by considering an aerodynamic and friction damping. The models were compared with the analytical results, and for the accurate estimation of the FRF magnitudes peak obtained via SEM, we used the Wittrick-Williams algorithm. Also, local sensitivity analyses were performed with respect to the tensile load and damping parameters. For the higher tensile load up to 10^4 kgf, the dispersion diagram of the conductor generated non-dispersive waves, and for tensile load below 10^3 kgf dispersive waves were observed. Such behaviour was followed by dynamic responses. The numerical FRFs calculated using both numerical models were compared with experimental FRFs measured at the laboratory span of the Electric Power Research Center (CEPEL). Therefore, the computational SEM models taking into account dissipative mechanisms associated to the cable dynamics provided accurate predictions within the measured data domain.

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